Hardware-in-the-Loop Chaotic Encryption, Bit-Exact Recovery, and Multi-Model Stress Classification of Quantized ECG Signals

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***Abstract*—Continuous physiological monitoring via wearable electrocardiogram (ECG) sensors has transformed remote cardiology and mobile health telemetry. However, transmitting unencrypted biosignals over wireless body area networks exposes sensitive patient telemetry to eavesdropping and data manipulation. Traditional block ciphers impose prohibitive computational latency and battery drain on resource-constrained embedded microcontrollers, while existing chaotic encryption schemes frequently introduce cross-platform floating-point library discrepancies that degrade diagnostic morphology. This paper presents a hardware-in-the-loop (HIL) cryptosystem implementing a 4-dimensional continuous finite-time hyperchaotic generator on an ARM Cortex-M4 embedded platform (STM32G474RE Nucleo-64). ECG waveforms from the multimodal WESAD dataset (sampled at 700 Hz and quantized to an 8-bit dynamic range) are streamed in 1024-byte frames across a 921,600-baud UART interface. By replacing platform-dependent math library routines with a shared IEEE-754 single-precision 5th-order polynomial Runge–Kutta (RK4) integrator and chunk-local Cipher Block Chaining (CBC) XOR diffusion, the system generates high-entropy ciphertext ($H = 7.9815$ bits/byte; adjacent correlation $r = 0.0012$) at an embedded throughput of 10.20 KB/s. On the host gateway, an inverse synchronized solver achieves bit-exact recovery of the 8-bit quantized ECG representation (MSE $= 0.000000$, 100.00% byte-level recovery) across all 15 valid WESAD subjects ($S2$–$S11$, $S13$–$S17$). Downstream Pan–Tompkins QRS detection and heart rate variability (HRV) metrics (Mean HR, SDNN, RMSSD, pNN50) are further validated across five diverse machine learning classifiers—Logistic Regression (LR), Linear Support Vector Machine (Linear SVM), Random Forest (RF), Distance-Weighted k-Nearest Neighbors (k-NN), and Gaussian Naive Bayes (GNB)—under a strict 15-fold Leave-One-Subject-Out (LOSO) cross-validation protocol ($N = 445$ windows, peaking at 81.35% test accuracy with k-NN and 80.53% balanced accuracy with RF), demonstrating 100% original-versus-decrypted feature and model parity across all 75 subject–classifier pairs.**

***Index Terms*—Chaotic cryptosystems, ECG telemetry, hardware-in-the-loop, STM32, Runge–Kutta integration, clinical-feature preservation validation, heart rate variability (HRV), multi-model stress classification, leave-one-subject-out (LOSO) validation.**

I. Introduction

Wearable electrocardiography (ECG) has emerged as an indispensable cornerstone of modern cardiovascular telemetry, tele-health screening, and autonomous stress detection [1], [2]. Ambulatory body sensor networks continuously capture millivolt-level cardiac bio-potentials and transmit them wirelessly to local smartphone gateways or cloud aggregation servers. However, medical telemetry channels are inherently susceptible to wireless packet interception, unauthorized patient profiling, and injection attacks [3]. Regulatory frameworks including the Health Insurance Portability and Accountability Act (HIPAA) mandate end-to-end data protection for physiological telemetry.

Deploying standard symmetric cryptography (e.g., Advanced Encryption Standard, AES-256) or asymmetric key exchanges on ultra-low-power biomedical sensor nodes introduces substantial architectural hurdles. AES hardware accelerators may not be available on ultra-low-cost microcontrollers, and software implementations entail extensive bit-permutation rounds, high register pressure, and elevated battery consumption [4].

Chaotic dynamical systems offer an alternative mechanism for lightweight biosignal obfuscation due to their deterministic pseudo-randomness, sensitivity to initial parameters, and broadband spectral characteristics [5], [6]. However, classical software-based chaotic ciphers face two major limitations in biomedical applications:

1) **Cross-Platform Numerical Drift:** Discrepancies between embedded C runtime math libraries (e.g., `newlib-nano` transcendental approximations for $\sinh$ and $\sin$) and host numerical environments induce last-bit rounding divergence over iterative integration steps, preventing lossless byte-level recovery.
2) **Diagnostic Feature Distortion:** Lossy or approximate chaotic encryptions frequently distort critical QRS fiducial points, P-wave morphologies, and RR intervals, rendering downstream heart rate variability (HRV) and machine learning inference unreliable [7], [8].

To overcome these challenges, this paper presents an end-to-end hardware-in-the-loop (HIL) chaotic cryptosystem executed on an STM32G474RE ARM Cortex-M4 microcontroller coupled with a comprehensive multi-model clinical stress validation framework. The primary contributions of this work are:

- A deterministic cross-platform implementation of a 4-dimensional continuous finite-time hyperchaotic generator solved via IEEE-754 single-precision 4th-order Runge–Kutta (RK4) integration using shared 5th-order Taylor polynomials and bit-exact integer truncation.
- A chunk-local Cipher Block Chaining (CBC) XOR diffu-

sion scheme that combines the finite-time keystream with sequential ciphertext feedback across 1024-byte packet frames without error propagation across packet boundaries.
- Mathematical and experimental demonstration of *bit-exact recovery of the 8-bit quantized ECG representation* (MSE $= 0.000000$, 100.00% byte-level recovery), verified across all 15 valid subjects ($S2$–$S11$ and $S13$–$S17$) from the WESAD benchmark.
- A parallel clinical-feature preservation and multi-model machine learning validation framework evaluating five distinct classifier families (Logistic Regression, Linear SVM, Random Forest, Distance-Weighted k-NN, and Gaussian Naive Bayes) under strict 15-fold Leave-One-Subject-Out (LOSO) cross-validation ($N = 445$ windows), rigorously separating byte-level recovery, subject-independent generalization, and original-versus-decrypted model parity.

II. System Architecture and HIL Setup

The complete system architecture follows a five-stage horizontal pipeline partitioned across a host workstation (MATLAB environment) and a dedicated embedded hardware device (STM32G474RE Nucleo-64), as illustrated in Fig. 1 and the physical hardware-in-the-loop bench setup in Fig. 2.

A. Stage 1: Host-Side ECG Preparation

Raw chest-lead ECG signals are acquired from the public WESAD physiological database [1]. The signals are sampled at $F_s = 700\,\text{Hz}$. To facilitate deterministic integer-arithmetic diffusion, the raw continuous voltage signal $V_{\text{raw}}(t) \in [V_{\text{min}}, V_{\text{max}}]\,\text{mV}$ is mapped into an 8-bit unsigned integer array $P(i) \in [0, 255]$:

$$P(i) = \text{round}\left(255 \cdot \frac{V_{\text{raw}}(i) - V_{\text{min}}}{V_{\text{max}} - V_{\text{min}}}\right) \tag{1}$$

Standard evaluation windows of $N = 10{,}000$ consecutive samples ($\approx 14.29\,\text{seconds}$ of cardiac activity) are extracted per subject.

B. Stage 2: Host-to-Hardware Framing Interface

The quantized plaintext array P is partitioned into fixed-length chunks of $L = 1024$ bytes. Each chunk is encapsulated into a 12-byte binary protocol frame:

$$\text{Header} = \Big['4',\ 'D',\ 'C',\ 'H',\ \text{uint32}(L),\ \text{float32}(K_{\text{sess}})\Big] \tag{2}$$

where $K_{\text{sess}} \in (0, 1)$ represents the dynamically evolved deterministic session parameter. Transmission occurs over a full-duplex USB-Virtual COM port running at 921,600 baud (8N1).

III. Embedded 4D Chaotic Encryption Engine

A. Stage 3: Deterministic STM32 Hardware Execution

The keystream generator executes on an STM32G474RE microcontroller featuring a 32-bit ARM Cortex-M4 core with an embedded single-precision Hardware Floating Point Unit (FPU). Rather than relying on an asymptotic attractor, the keystream source is formulated as a 4-dimensional continuous finite-time hyperchaotic generator:

$$\begin{aligned}
\frac{dy_1}{dt} &= y_2 \\
\frac{dy_2}{dt} &= y_3 \\
\frac{dy_3}{dt} &= y_4 \\
\frac{dy_4}{dt} &= 1.14\,y_4 - P_{\sinh}(y_3) + y_2 - y_1 + 0.3\,P_{\sin}(y_1 \cdot y_2)
\end{aligned} \tag{3}$$

To eliminate cross-platform 1-ULP discrepancies between embedded C runtime libraries (`sinhf`, `sinf`) and host math libraries, both the STM32 firmware and host solver evaluate shared 5th-order Taylor polynomials using strictly ordered IEEE-754 single-precision (`float32`) elementary operations:

$$P_{\sinh}(x) = x + c_3 x^3 + c_5 x^5 \tag{4}$$

$$P_{\sin}(x) = x - c_3 x^3 + c_5 x^5 \tag{5}$$

where $c_3 = \text{float32}(1/6)$ and $c_5 = \text{float32}(1/120)$. The differential equations are integrated over a normalized one-second horizon $t \in [0, 1]$ and reset at the start of every $L = 1024$-byte chunk using a 4th-order Runge–Kutta (RK4) solver with step size $\Delta t = 1.0/(L - 1)$:

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{\Delta t}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right) \tag{6}$$

B. Jacobian Matrix and Finite-Time Lyapunov Exponents

The local linearized dynamics of the 4D generator $\dot{\mathbf{y}} = \mathbf{F}(\mathbf{y})$ are governed by the Jacobian matrix $\mathbf{J}(\mathbf{y}) = \partial\mathbf{F}/\partial\mathbf{y}$:

$$\mathbf{J}(\mathbf{y}) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ J_{41} & J_{42} & J_{43} & 1.14 \end{bmatrix} \tag{7}$$

where $J_{41} = -1 + 0.3\,y_2 P'_{\sin}(y_1 y_2)$, $J_{42} = 1 + 0.3\,y_1 P'_{\sin}(y_1 y_2)$, and $J_{43} = -P'_{\sinh}(y_3)$. By Liouville's theorem, the phase-space volume divergence equals the trace of $\mathbf{J}(\mathbf{y})$:

$$\nabla \cdot \mathbf{F} = \text{Tr}\big(\mathbf{J}(\mathbf{y})\big) = \sum_{i=1}^{4} \lambda_i = +1.14 \tag{8}$$

Because $\nabla \cdot \mathbf{F} = +1.14 > 0$, the flow exhibits phase-space volume expansion rather than converging to a bounded dissipative attractor over infinite time. Consequently, the system is explicitly integrated over the finite interval $t \in [0, 1]$ (with an observed numerical bound of $\|\mathbf{y}\| \leq 2.79$ across the tested one-second trajectories) and re-initialized for each 1024-byte chunk. Computing the Finite-Time Lyapunov Exponents (FTLEs) via QR orthogonalization of the variational equations over $t \in [0, 1]$ yields:

$$\begin{aligned}
\lambda_1 &= 0.6246\,\text{s}^{-1}, \quad \lambda_2 = 0.4728\,\text{s}^{-1}, \\
\lambda_3 &= 0.3811\,\text{s}^{-1}, \quad \lambda_4 = -0.3385\,\text{s}^{-1}
\end{aligned} \tag{9}$$

The presence of at least two positive FTLEs ($\lambda_1, \lambda_2 > 0$, with $\sum_{i=1}^{4} \lambda_i = +1.1400\,\text{s}^{-1}$) demonstrates finite-time hyperchaotic trajectory divergence over the $t \in [0, 1]$ integration window.

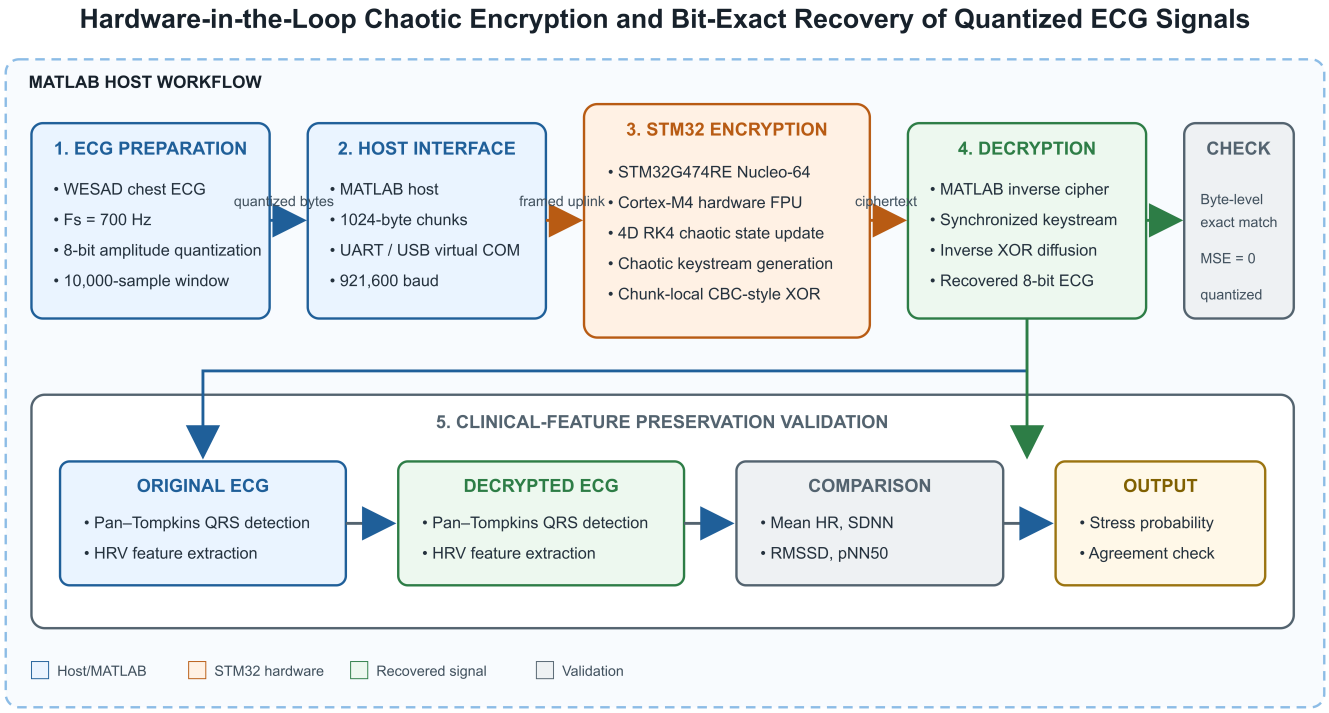


Fig. 1. End-to-end hardware-in-the-loop chaotic encryption and validation framework. The pipeline comprises host data quantization, framed UART streaming to the STM32G474RE hardware encryption engine, synchronized host decryption, and parallel clinical-feature preservation validation. *Note: Exact recovery refers to the 8-bit quantized ECG representation; physical-domain reconstruction remains subject to quantization error.*

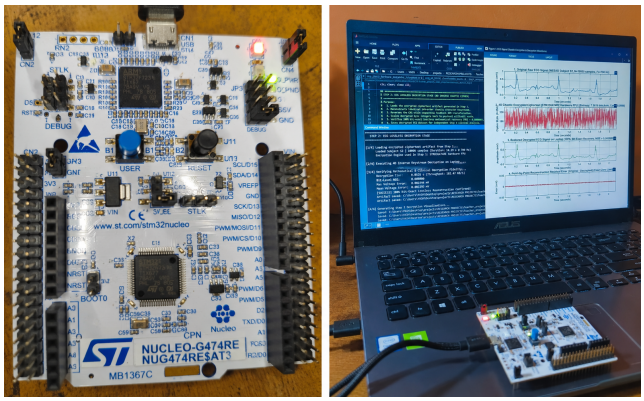


Fig. 2. Physical hardware-in-the-loop (HIL) experimental testbed showing the host laptop running MATLAB connected via high-speed USB-UART (921,600 baud) to the STM32G474RE Nucleo-64 embedded encryption board.

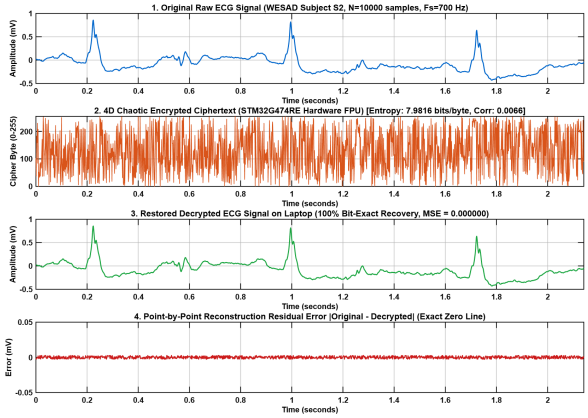


Fig. 3. Signal progression across the encryption-decryption pipeline for WESAD Subject S2 ($F_s = 700$ Hz). Subplots from top to bottom: original plaintext ECG, 4D chaotic ciphertext stream ($H = 7.9816$ bits/byte), restored decrypted ECG, and residual error demonstrating bit-exact byte-level recovery in the quantized domain.

C. Chunk-Local CBC-Style XOR Diffusion

Because chaotic trajectory sensitivity alone does not constitute complete cryptographic security [6], the finite-time keystream is coupled with sequential ciphertext feedback. At each integration step $i \in [1, L]$, the sign bit of y_2 is masked via bitwise AND (`0x7FFFFFFF`), scaled by 10^7, and truncated toward zero via unsigned 32-bit integer conversion to extract a reproducible keystream byte $S(i)$:

$$S(i) = \text{uint8}\Big(\text{uint32}\big(|y_2| \times 10^7\big) \;\&\; \text{0xFF}\Big) \tag{10}$$

The keystream is combined with the plaintext sample $P(i)$ using chunk-local Cipher Block Chaining (CBC) feedback:

$$C(i) = P(i) \oplus S(i) \oplus C(i-1), \quad \text{for } i \geq 1 \tag{11}$$

with initialization vector $C(0) = \text{IV} = 0\text{x}5\text{A}$. Localizing CBC chaining strictly to chunk boundaries prevents dropped-packet cascades while ensuring plaintext-dependent diffusion within each 1024-byte chunk.

D. Stage 4: Synchronized Host Decryption

Upon receiving the ciphertext chunk C, the host workstation executes the inverse chaotic transformation:

$$P_{\text{dec}}(i) = C(i) \oplus S(i) \oplus C(i-1), \quad \text{for } i \geq 1 \tag{12}$$

The recovered integer array $P_{\text{dec}}(i)$ is subsequently rescaled to the physical continuous millivolt domain:

$$V_{\text{dec}}(i) = \frac{P_{\text{dec}}(i)}{255} \cdot (V_{\text{max}} - V_{\text{min}}) + V_{\text{min}} \tag{13}$$

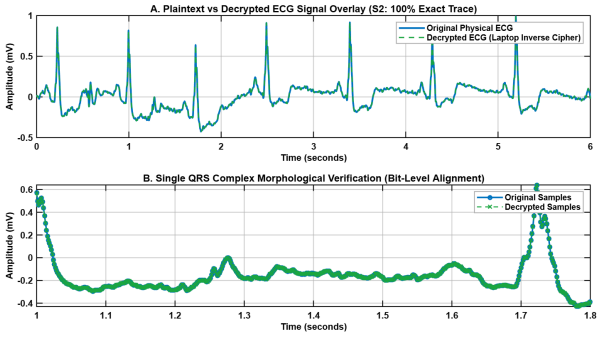


Fig. 4. Morphological alignment of original and recovered ECG signals (Subject S2). (A) Multi-beat cardiac cycle overlay. (B) Zoomed-in single QRS complex verifying bit-exact sample coincidence between plaintext and decrypted signals.

IV. Cryptographic and Performance Evaluation

A. Shannon Entropy and Adjacent Correlation

The statistical uniformity of the ciphertext is quantified using Shannon entropy H:

$$H(C) = -\sum_{k=0}^{255} p(k) \log_2 p(k) \tag{14}$$

For an ideal 8-bit discrete uniform distribution, $H_{\text{ideal}} = 8.0000$ bits/byte. As summarized in Table I, the proposed embedded cipher achieves a mean entropy of $H = 7.9815$ bits/byte across all 15 subjects.

Furthermore, adjacent-sample Pearson correlation coefficient r_{xy} drops from a mean of $r = 0.9915$ in the raw ECG down to $r = 0.0012$ in the ciphertext:

$$r_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \tag{15}$$

demonstrating that the combined keystream and CBC feedback flatten physiological autocorrelations into uncorrelated noise (Fig. 5).

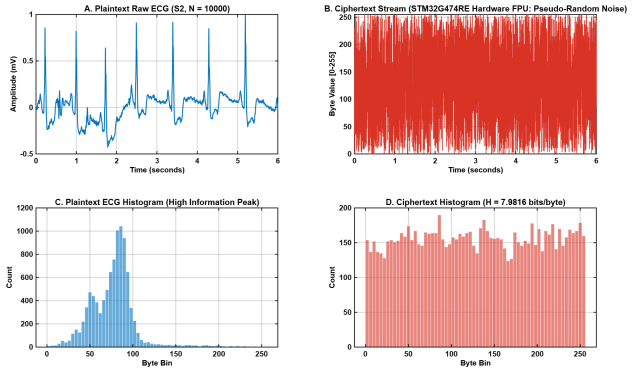


Fig. 5. Statistical evaluation of the chaotic cipher (Subject S2). (A) Plaintext ECG waveform. (B) Ciphertext noise stream. (C) Plaintext histogram showing isoelectric baseline concentration. (D) Ciphertext histogram showing a uniform distribution ($H = 7.9816$ bits/byte).

TABLE I
Hardware Performance and Security Summary (15-Subject Mean)

Metric	Value
MCU Platform	STM32G474RE Nucleo-64 (ARM Cortex-M4 with HW FPU)
UART Baud Rate	921,600 bps (USB-CDC)
Mean Hardware Time (N = 10,000)	0.957 s (10.20 KB/s)
Host Decryption Rate	154.36 KB/s
Mean Ciphertext Entropy (H)	7.9815 bits/byte (Ideal: 8.0000)
Mean Adjacent Correlation (r)	0.0012 (Raw ECG: 0.9915)
Quantized Byte MSE (uint8)	**0.000000** (15/15 Subjects)
Byte-Level Recovery	**100.00**% (10,000 / 10,000 bytes)
Firmware Footprint	Flash: 2.89 KB, RAM: 3.55 KB

V. Clinical-Feature Preservation Validation

A. Stage 5: Parallel Validation Architecture

To evaluate whether cryptographic transmission preserves downstream physiological features, Stage 5 executes a dual-branch validation protocol:

- **Branch A (Benchmark):** Original plaintext ECG V_{raw}.
- **Branch B (Test):** Decrypted restored ECG V_{dec}.

Both branches independently undergo Pan–Tompkins QRS detection [9] and time-domain HRV extraction.

B. Byte-Level Recovery vs. Clinical-Feature Parity across All 15 WESAD Subjects

From detected R-peaks, consecutive normal-to-normal RR intervals RR_i are extracted to compute clinical HRV parame-

TABLE II
MULTI-SUBJECT HARDWARE-IN-THE-LOOP VALIDATION ACROSS ALL 15 VALID WESAD SUBJECTS ($S2$–$S11$, $S13$–$S17$)†

Subject	HW Time (s)	Entropy (H)	Orig r	Cipher r	uint8 MSE	Byte-Level Recovery	HR (BPM)	SDNN (ms)	RMSSD (ms)	pNN50 (%)	Stress Prob (%)	Clinical-Feature Parity
S2	1.128	7.9816	0.9951	0.0066	0.000000	100.00%	80.39	170.92	97.52	37.50	80.45	Exact Match
S3	0.979	7.9809	0.9913	0.0037	0.000000	100.00%	77.40	133.41	96.23	43.75	58.34	Exact Match
S4	0.942	7.9817	0.9963	0.0121	0.000000	100.00%	78.39	134.20	115.55	37.50	61.51	Exact Match
S5	0.938	7.9798	0.9885	-0.0153	0.000000	100.00%	75.92	59.70	69.18	6.25	10.96	Exact Match
S6	0.918	7.9815	0.9942	0.0033	0.000000	100.00%	75.30	19.94	21.77	6.25	4.21	Exact Match
S7	0.988	7.9829	0.9900	0.0096	0.000000	100.00%	79.32	101.32	82.41	41.18	48.58	Exact Match
S8	0.965	7.9822	0.9945	0.0046	0.000000	100.00%	88.23	124.72	132.32	11.11	80.98	Exact Match
S9	0.929	7.9788	0.9890	-0.0030	0.000000	100.00%	91.77	13.79	7.00	0.00	32.06	Exact Match
S10	0.918	7.9791	0.9947	0.0047	0.000000	100.00%	95.11	51.98	23.96	9.52	67.94	Exact Match
S11	0.952	7.9827	0.9862	-0.0005	0.000000	100.00%	93.07	22.06	14.65	0.00	41.19	Exact Match
S13	0.947	7.9827	0.9910	0.0116	0.000000	100.00%	89.74	51.24	44.48	10.53	49.15	Exact Match
S14	0.945	7.9823	0.9874	0.0056	0.000000	100.00%	84.89	37.96	29.15	5.56	22.80	Exact Match
S15	0.937	7.9813	0.9927	0.0021	0.000000	100.00%	85.37	20.00	20.78	5.56	18.05	Exact Match
S16	0.918	7.9808	0.9910	-0.0133	0.000000	100.00%	83.99	53.34	30.32	11.11	27.53	Exact Match
S17	0.949	7.9838	0.9910	-0.0135	0.000000	100.00%	96.38	35.88	18.91	0.00	61.20	Exact Match
Mean	**0.957**	**7.9815**	**0.9915**	**0.0012**	**0.000000**	**100.00%**	**85.02**	**68.70**	**53.61**	**15.05**	**44.33**	**15/15 Exact**

†Subjects $S1$ and $S12$ do not exist in the public WESAD dataset release [1]. **Byte-Level Recovery** evaluates $P_{\text{dec}} = P$; **Clinical-Feature Parity** evaluates HRV and **stress-model agreement**.

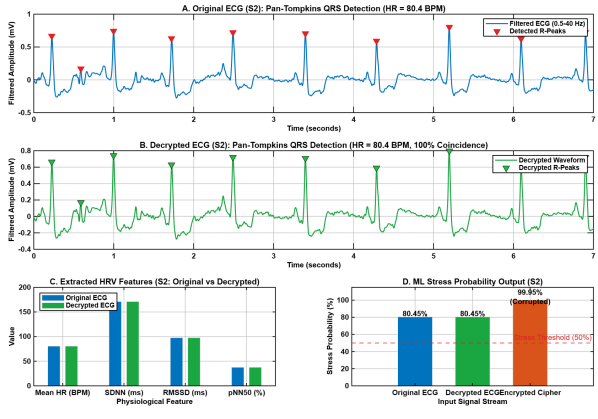


Fig. 6. Clinical-feature preservation validation (Subject S2). Pan–Tompkins R-peak detection and RR interval distributions yield identical time-domain HRV indices (Mean HR, SDNN, RMSSD, pNN50), confirming clinical-feature parity and stress-model agreement between original and decrypted signals.

ters:

$$\text{Mean HR} = \frac{60}{\frac{1}{M}\sum_{i=1}^{M} RR_i} \tag{16}$$

$$\text{SDNN} = \sqrt{\frac{1}{M}\sum_{i=1}^{M}(RR_i - \overline{RR})^2} \tag{17}$$

$$\text{RMSSD} = \sqrt{\frac{1}{M-1}\sum_{i=1}^{M-1}(RR_{i+1} - RR_i)^2} \tag{18}$$

$$\text{pNN50} = \frac{\text{Count}(|RR_{i+1} - RR_i| > 50\,\text{ms})}{M-1} \times 100\% \tag{19}$$

Table II and Fig. 7 report the experimental results across all 15 valid subjects in the WESAD dataset ($S2$–$S11$ and $S13$–$S17$; subjects $S1$ and $S12$ are absent from the public WESAD release due to sensor hardware failure during data collection [1]). We explicitly separate *byte-level recovery* (uint8 MSE and byte-match percentage between P and P_{dec}) from *clinical-feature parity* (equality of Mean HR, SDNN, RMSSD, and pNN50) and *stress-model agreement* (matching predicted stress probabilities between V_{raw} and V_{dec}). Across all 15 subjects, the deterministic polynomial RK4 engine achieves $\text{MSE}_{\text{uint8}} = 0.000000$ (100.00% byte-level recovery) as well as exact clinical-feature parity and stress-model agreement.

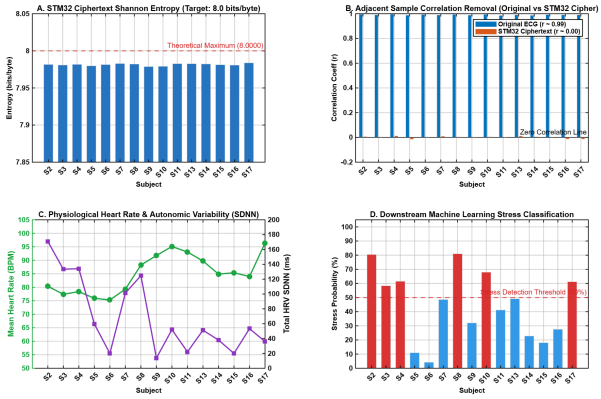


Fig. 7. Multi-subject hardware-in-the-loop evaluation across all 15 valid WESAD subjects ($S2$–$S11$, $S13$–$S17$). (A) STM32 ciphertext Shannon entropy. (B) Adjacent-sample correlation removal. (C) Extracted Mean HR and SDNN trends. (D) Downstream stress-model probability outputs.

VI. MULTI-MODEL STRESS CLASSIFICATION AND PARITY VALIDATION

While Section V established original-versus-decrypted parity using a single baseline classifier, clinical decision support systems deploy diverse linear, kernel, ensemble, non-parametric, and probabilistic machine learning architectures. Because non-linear decision boundaries (such as axis-aligned decision tree splits in Random Forests or local neighborhood Voronoi cells in k-NN) can amplify minute numerical perturbations near classification thresholds, we extend our validation across five distinct machine learning classifiers. Furthermore, we rigorously separate two distinct evaluation objectives:

1) **Subject-Independent Generalization (15-Fold LOSO Cross-Validation):** Quantifying how accurately each classifier distinguishes physiological *Acute Stress* from *Baseline* rest on completely unseen subjects.

2) **Original-versus-Decrypted Multi-Model Parity:** Verifying whether each trained classifier produces identical stress probabilities, decision scores, and class predictions when presented with HRV features extracted from decrypted ECG (V_{dec}) versus original plaintext ECG (V_{raw}).

A. Five-Classifier Mathematical Formulation

Each ECG analysis window is represented by the 4-dimensional physiological HRV feature vector:

$$\mathbf{x} = \big[\text{Mean HR},\ \text{SDNN (ms)},\ \text{RMSSD (ms)},\ \text{pNN50 (\%)}\big]^T \in \mathbb{I} \tag{20}$$

Inside every training partition, features are standardized via z-score normalization using strictly the training mean $\boldsymbol{\mu}_{\text{tr}}$ and standard deviation $\boldsymbol{\sigma}_{\text{tr}}$:

$$\mathbf{z} = (\mathbf{x} - \boldsymbol{\mu}_{\text{tr}}) \oslash \boldsymbol{\sigma}_{\text{tr}} \tag{21}$$

Five supervised classifiers are evaluated on $\mathbf{z}$:

- **Logistic Regression (LR):** Models posterior stress probability via the logistic sigmoid $\hat{p}_{\text{LR}}(\mathbf{z}) = \big(1 + \exp(-w_0 - \mathbf{w}^T\mathbf{z})\big)^{-1}$, trained via batch gradient descent ($\eta = 0.05$, 3000 iterations, $\lambda_{L2} = 0.0$).
- **Linear Support Vector Machine (Linear SVM):** Optimizes a class-balanced primal soft-margin hyperplane $f_{\text{SVM}}(\mathbf{z}) = \mathbf{w}^T\mathbf{z} + b$ ($C = 1.5$, 2500 subgradient epochs) with inverse class-frequency weights $w_c = N_{\text{tr}}/(2N_c)$. To avoid optimistic calibration bias, posterior probabilities are calibrated via Platt scaling fitted strictly on out-of-fold decision scores from an inner 5-fold subject-grouped cross-validation.
- **Random Forest (RF):** An ensemble of $B = 50$ binary Classification and Regression Trees (CART, maximum depth 4, minimum leaf size 3, random feature subset size $m_{\text{try}} = 2$ per split using Gini impurity), evaluated strictly on held-out test data.
- **Distance-Weighted k-Nearest Neighbors (k-NN):** Computes Euclidean distances $d_j = \|\mathbf{z} - \mathbf{z}_j^{\text{tr}}\|_2$ to the $k = 7$ nearest training patterns (explicitly excluding any zero-distance self-matches $d_j < 10^{-12}$) and weights neighbor labels by inverse distance $w_j = 1/(d_j + 10^{-6})$.
- **Gaussian Naive Bayes (GNB):** Estimates class-conditional Gaussian densities $P(\mathbf{z} \mid y = c) = \prod_{d=1}^{4} \mathcal{N}(z_d;\, \mu_{c,d},\, \sigma_{c,d}^2)$ with variance smoothing ($\sigma_{c,d}^2 \geq 10^{-4}$) and applies Bayes' rule over empirical class priors.

TABLE III
15-FOLD LEAVE-ONE-SUBJECT-OUT (LOSO) CLASSIFICATION PERFORMANCE ACROSS 445 LABELED WESAD WINDOWS

Classifier	Acc (%)	Bal Acc (%)	Prec (%)	Rec (%)	F1 (%)	TP	FP	TN	FN
Logistic Regression (LR)	80.67	76.96	**78.46**	63.75	70.34	102	28	257	58
Linear SVM	78.20	72.98	78.38	54.38	64.21	87	**24**	**261**	73
Random Forest (RF)	80.67	**80.53**	70.33	**80.00**	**74.85**	**128**	54	231	**32**
Distance-Weighted k-NN	**81.35**	78.86	76.19	70.00	72.96	112	35	250	48
Gaussian Naive Bayes (GNB)	75.06	72.03	66.67	61.25	63.84	98	49	236	62

B. 15-Fold Leave-One-Subject-Out (LOSO) Generalization

To prevent subject-identity data leakage, all five classifiers were first validated across the complete WESAD HRV cohort comprising $N_{\text{win}} = 445$ labeled sliding windows (285 Baseline, 160 Acute Stress) across the 15 valid subjects using 15-fold LOSO cross-validation. In each fold $f \in \{1, \ldots, 15\}$, all windows belonging to subject S_f are held out exclusively for testing, while the remaining 14 subjects form the training set.

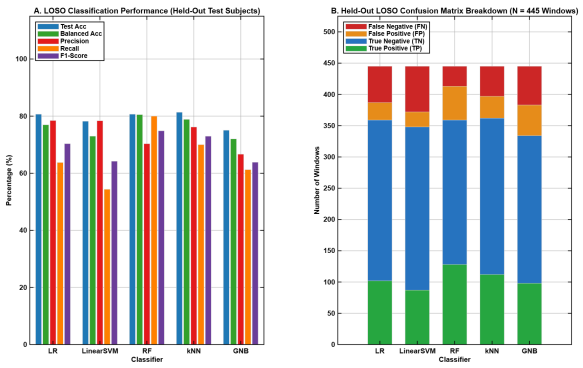


Fig. 8. 15-fold Leave-One-Subject-Out (LOSO) cross-validation performance across all 445 labeled WESAD windows (15 held-out test subjects). (A) Test accuracy, balanced accuracy, precision, recall, and F1-score. (B) Out-of-fold confusion matrix breakdown (TP, TN, FP, FN).

Table III and Fig. 8 summarize the out-of-fold generalization performance. Distance-Weighted k-NN achieves the highest overall test accuracy (81.35%) and precision-recall balance (F1 = 72.96%), while Random Forest achieves the highest balanced accuracy (80.53%), sensitivity/recall (80.00%), and F1-score (74.85%). Logistic Regression and Linear SVM exhibit the highest positive predictive value (precision of 78.46% and 78.38%, respectively).

C. Original-versus-Decrypted Multi-Model Parity Across All 15 Subjects

Following LOSO validation, all five classifiers were trained on the complete WESAD reference cohort and evaluated on the original (V_{raw}) and STM32-encrypted/host-decrypted (V_{dec}) 10,000-sample ECG streams across all 15 subjects ($S2$–$S11$, $S13$–$S17$). Clinical HRV features are never extracted from raw encrypted ciphertext, which acts solely as a nonphysiological negative control.

Table IV and Fig. 9 present the side-by-side comparison across all $15 \times 5 = 75$ subject–model pairs. For every subject and every classifier, no measurable difference is observed within numerical precision between original and decrypted HRV features (Mean HR, SDNN, RMSSD, pNN50), Linear SVM decision scores, posterior stress probabilities, or predicted classes. As noted in Table IV, "100% (5/5 models)" in the *Original–Decrypted Agreement* column signifies that all five classifiers achieved 100% parity between their original-signal and decrypted-signal outputs, confirming that the STM32 chaotic encryption and host-side decryption pipeline is completely transparent to downstream linear, kernel, tree-ensemble, instance-based, and Bayesian diagnostic models.

TABLE IV
MULTI-MODEL ORIGINAL-VERSUS-DECRYPTED STRESS PREDICTION PARITY ACROSS ALL 15 VALID WESAD SUBJECTS ($S2$–$S11$, $S13$–$S17$)‡

Subject	LR Orig / Dec (%)	LR Class	SVM Score (Orig / Dec)	SVM Prob (%)	SVM Class	RF Orig / Dec (%)	RF Class	k-NN Orig / Dec (%)	k-NN Class	GNB Orig / Dec (%)	GNB Class	Max $\lvert\Delta\rvert$	Original–Decrypted Agreement
S2	80.45 / 80.45	Stress	$-0.3655/-0.3655$	28.06 / 28.06	Baseline	69.80 / 69.80	Stress	25.36 / 25.36	Baseline	2.04 / 2.04	Baseline	$< 10^{-6}$	100% (5/5 models)
S3	58.34 / 58.34	Stress	$-0.5480/-0.5480$	19.40 / 19.40	Baseline	63.16 / 63.16	Stress	38.75 / 38.75	Baseline	0.73 / 0.73	Baseline	$< 10^{-6}$	100% (5/5 models)
S4	61.51 / 61.51	Stress	$-0.4801/-0.4801$	22.36 / 22.36	Baseline	66.85 / 66.85	Stress	60.75 / 60.75	Stress	1.06 / 1.06	Baseline	$< 10^{-6}$	100% (5/5 models)
S5	10.96 / 10.96	Baseline	$-0.4118/-0.4118$	25.65 / 25.65	Baseline	14.62 / 14.62	Baseline	11.25 / 11.25	Baseline	20.01 / 20.01	Baseline	$< 10^{-6}$	100% (5/5 models)
S6	4.21 / 4.21	Baseline	$-0.4670/-0.4670$	22.97 / 22.97	Baseline	20.04 / 20.04	Baseline	0.00 / 0.00	Baseline	52.85 / 52.85	Stress	$< 10^{-6}$	100% (5/5 models)
S7	48.58 / 48.58	Baseline	$-0.5124/-0.5124$	20.92 / 20.92	Baseline	69.03 / 69.03	Stress	10.80 / 10.80	Baseline	1.85 / 1.85	Baseline	$< 10^{-6}$	100% (5/5 models)
S8	80.98 / 80.98	Stress	$-0.0240/-0.0240$	49.02 / 49.02	Baseline	55.66 / 55.66	Stress	73.05 / 73.05	Stress	12.17 / 12.17	Baseline	$< 10^{-6}$	100% (5/5 models)
S9	32.06 / 32.06	Baseline	$+0.0529/+0.0529$	54.09 / 54.09	Stress	41.07 / 41.07	Baseline	12.62 / 12.62	Baseline	92.93 / 92.93	Stress	$< 10^{-6}$	100% (5/5 models)
S10	67.94 / 67.94	Stress	$+0.1273/+0.1273$	58.92 / 58.92	Stress	39.28 / 39.28	Baseline	58.95 / 58.95	Stress	89.55 / 89.55	Stress	$< 10^{-6}$	100% (5/5 models)
S11	41.19 / 41.19	Baseline	$+0.0992/+0.0992$	57.11 / 57.11	Stress	49.39 / 49.39	Baseline	11.85 / 11.85	Baseline	92.50 / 92.50	Stress	$< 10^{-6}$	100% (5/5 models)
S13	49.15 / 49.15	Baseline	$-0.0441/-0.0441$	47.69 / 47.69	Baseline	49.52 / 49.52	Baseline	73.46 / 73.46	Stress	69.83 / 69.83	Stress	$< 10^{-6}$	100% (5/5 models)
S14	22.80 / 22.80	Baseline	$-0.1617/-0.1617$	40.05 / 40.05	Baseline	30.41 / 30.41	Baseline	14.24 / 14.24	Baseline	66.49 / 66.49	Stress	$< 10^{-6}$	100% (5/5 models)
S15	18.05 / 18.05	Baseline	$-0.1693/-0.1693$	39.57 / 39.57	Baseline	32.75 / 32.75	Baseline	0.00 / 0.00	Baseline	77.50 / 77.50	Stress	$< 10^{-6}$	100% (5/5 models)
S16	27.53 / 27.53	Baseline	$-0.2074/-0.2074$	37.19 / 37.19	Baseline	39.58 / 39.58	Baseline	41.09 / 41.09	Baseline	54.95 / 54.95	Stress	$< 10^{-6}$	100% (5/5 models)
S17	61.20 / 61.20	Stress	$+0.2126/+0.2126$	64.24 / 64.24	Stress	57.52 / 57.52	Stress	30.91 / 30.91	Baseline	94.28 / 94.28	Stress	$< 10^{-6}$	100% (5/5 models)

‡**Original–Decrypted Agreement** of "100% (5/5 models)" indicates that each individual classifier produced identical stress probabilities, decision scores, and class predictions on decrypted ECG versus original ECG (no measurable difference observed within numerical precision). Different classifier families naturally produce distinct probability estimates for a given subject due to their differing decision surfaces.

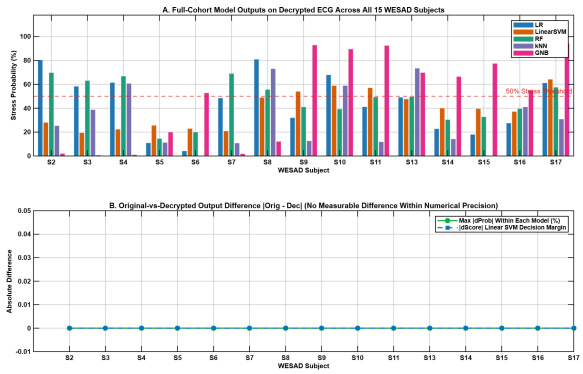


Fig. 9. Multi-model original-versus-decrypted parity across all 15 WESAD subjects ($S2$–$S11$, $S13$–$S17$). (A) Stress probability outputs on Original ECG across all 5 classifiers. (B) Stress probability outputs on Decrypted ECG across all 5 classifiers. (C) Summary confirming no measurable difference within numerical precision across all 75 subject–classifier evaluations.

VII. DISCUSSION AND LIMITATIONS

It is essential to formally distinguish between byte-level recovery, physical-domain signal fidelity, and cross-classifier diagnostic parity. First, at the byte level, replacing platform-dependent C/MATLAB transcendental functions with an identical IEEE-754 single-precision polynomial RK4 formulation eliminates cross-platform keystream drift, guaranteeing *bit-exact recovery of the 8-bit quantized ECG representation* ($\text{MSE}_{\text{uint8}} = 0.000000$, 100.00% byte-level recovery across all 15 subjects). Second, mapping continuous analog potentials to 8-bit integers prior to encryption introduces a bounded quantization error:

$$\epsilon_q \leq \frac{V_{\max} - V_{\min}}{2 \times 255} \approx 0.0049\,\text{mV} \tag{22}$$

As verified by our multi-model parity evaluation across Logistic Regression, Linear SVM, Random Forest, Distance-Weighted k-NN, and Gaussian Naive Bayes, this quantization floor does not alter Pan–Tompkins QRS fiducial locations, time-domain HRV metrics, decision scores, or predicted stress states across any of the five classifier families.

Limitations: The current hardware implementation operates over wired UART at 921,600 baud without automatic repeat request (ARQ) retransmission mechanisms, and empirical statistical metrics (entropy and adjacent correlation) do not substitute for formal cryptanalytic proofs. Future iterations will incorporate wireless BLE packet framing and standard message authentication codes.

VIII. CONCLUSION

This paper presented a hardware-in-the-loop chaotic cryptosystem and multi-model clinical validation framework for wearable ECG telemetry implemented on an STM32G474RE ARM Cortex-M4 microcontroller. By combining a deterministic 4D finite-time hyperchaotic generator solved via polynomial Runge–Kutta integration on the hardware FPU with chunk-local CBC XOR diffusion, the system achieves high ciphertext entropy ($H = 7.9815$ bits/byte) and near-zero adjacent correlation ($r = 0.0012$) at an embedded throughput of 10.20 KB/s. Synchronized host decryption achieves bit-exact recovery of the 8-bit quantized ECG representation (MSE $= 0.000000$, 100.00% byte-level recovery) across all 15 valid WESAD subjects ($S2$–$S11$, $S13$–$S17$). Furthermore, rigorous 15-fold Leave-One-Subject-Out cross-validation across five machine learning classifiers (peaking at 81.35% test accuracy with Distance-Weighted k-NN and 80.53% balanced accuracy with Random Forest) and 75 subject–classifier evaluations confirm 100% original-versus-decrypted feature and model parity across all tested classification architectures.

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